Chapter 2

Materials for Electronic Applications

2.1 Nanomaterials

Introduction Nanomaterials are materials whose characteristic length scale lies within the nanometric range (1-100 nm) at least in one dimension. The word 'nano' has the origin from the Greek meaning 'Dwarf'. It is one billionth of a metre ($1/10^9$ m). Nanomaterials are objects with any one of the three external dimensions at the nanoscale. Nanoparticles that are naturally occurring (eg; volcanic ash, soot from forest fires) or are the incidental byproducts of combustion processes (eg; welding, diesel engines) are usually physically and chemically heterogeneous and often termed ultrafine particles. Engineered nanomaterials are intentionally produced and designed with very specific properties related to shape, size, surface properties and chemistry. These properties are reflected in aerosols, colloids, or powders.

The behavior of nanomaterials may depend more on surface area than particle composition itself. Two principal factors cause the properties of nanomaterials to differ significantly from other materials: increased relative surface area, and quantum effects. These factors can change or enhance properties such as reactivity, strength and electrical characteristics. As a particle decreases in size, a greater proportion of atoms are found at the surface compared to those inside. For example, a particle of size 30 nm has 5 percentage of its atoms on its surface, at 10 nm 20 percentage of its atoms, and at 3 nm 50 percentage of its atoms. Thus nanomaterials have a much greater surface area per unit mass compared with larger particles. As growth and catalytic chemical reactions occur at surfaces, this means that a given mass of material in nanoparticulate form will be much more reactive than the same mass of material made up of larger particles.

2.1.1 Classification

There are several ways of classification of nanomaterials. Here we discuss two types.

1. **Classification based on dimension:** This is the classification based on the number of dimensions, which are not confined to the nanoscale range (< 100 nm).

   (a) **Zero dimension (0-D):** Here all the three dimensions are in the nanometric range eg; Nano particles.

   (b) **One dimension (1-D):** Here one of the dimensions is outside the nanometric range and the other two are within the range eg; Nano wires, fibers and tubes.

(c) **Two dimension (2-D):** Here two of the dimensions are outside the nanometric range and one is within the range eg; Nano films, layers and coat.

(d) **Three dimension (3-D):** Here all the dimensions are outside the nanometric range (> 100 nm) eg; Bundles of nano wires and tubes, multinano layers.

2. **Classification based on materials**

(a) **Carbon based nanomaterials:** These are defined as materials in which the nanocomponent is pure carbon eg; Carbon nano tubes (CNT), wires, spheres (fullerenes) and grephene.

(b) **Metal based nanomaterials:** Metal-based nanomaterials are materials made of metallic nanoparticles like gold, silver, metal oxides, etc. For example, titanium dioxide (TiO_2).

(c) **Nanocomposites:** Composite nanomaterials contain a mixture of simple nanoparticles or compounds such as nanosized clays within a bulk material. The nanoparticles give better physical, mechanical, and/or chemical properties to the initial bulk material.

(d) **Nano polymers or Dendrimers:** Dendrimers are nanosized polymers built from branched units. These are tree-like molecules with defined cavities. They can be functionalized at the surface and can hide molecules in their cavities. A direct application of dendrimers is for drug delivery.

(e) **Biological nanomaterials:** These nanomaterials are of biological origin and are used for nanotechnological applications. The important features of these particles are i) self assembly properties and ii) specific molecular recognition eg; DNA nano particles, nanostructured peptides. Various self assembled peptide structures can be designed to release compounds under specific conditions and are used in drug delivery systems.

2.1.2 Synthesis of nanoparticles

There are many methods available for synthesis of nano-particles. Physical methods include Laser ablation, spluttering techniques etc. Some chemical methods are discussed below

Chemical synthesis of nano particles

1. **Hydrolysis:** Nano particles of metal oxides can be prepared by the hydrolysis of their alkoxide solutions under controlled conditions. Commercially important nano particles of silica (SiO_2),titania (TiO_2), alumina (Al_2O_3) are prepared by this method. Hydrothermal method and sol-gel method come under this category.

$$Ti(OR)_4 + 2H_2O \rightarrow TiO_2 + 4ROH$$

Sol-Gel method: The sol-gel method is based on the phase transformation of a sol into a gel. A sol is colloidal system of nano-solid particles dispersed in a liquid. A gel is colloidal system in which liquid droplets are dispersed in a network of solid nanoparticles. Hydrolysis of metallic alkoxides or metal salts can give a sol at a suitable temperature and pH. The sol contains many other impurities. In order to remove impurities sol is transformed into a gel by changing the pH or other factors. The gel can be purified by filtration and washing with suitable solvents. The purified gel on drying give solid nanoparticle. For example aluminium oxide nano particles are obtained by hydrolysis of aluminium alkoxide by sol-gel technique.

$$Al(OR)_3 + 3H_2O \rightarrow Al(OH)_3 + 3ROH$$

2. **Reduction:** Nano particles of gold and silver can be prepared by the reduction of their respective solutions using reducing agents, such as sodium borohydride, ascorbic acid, glucose etc. along with a protective agent like thyol, glucose etc. This method can be divided into two, reduction using reducing agents and electro reduction.

$$Ag^+ + e^- \rightarrow Ag$$

Reduction using reducing agents: Silver nano particles can be prepared by the following method. 60 mL of 1 mM $AgNO_3$ solution is taken in a beaker, covered with a watch glass and heated in hot plate. The solution is then stirred using a magnetic stirrer. On boiling the solution, 6mL of 10 mM of trisodium citrate is added dropwise, about one drop per second. The beaker is then closed and kept for some time till the colour of the solution changed to a light golden colour. Then it is allowed to cool. The solvent can be removed by freeze-drying.

Properties of Nanomaterials

1. **Physical Properties:** Crystal structure of nanoparticles is same as bulk structure with different lattice parameters. The inter-atomic spacing decreases with size and this is due to long range electrostatic forces and the short range core-core repulsion. The melting point of nanoparticles decreases with size.

2. **Chemical Properties:** A large fraction of the atoms are located at the surface of the nanomaterial which increase its reactivity and catalytic activity. Properties of materials with nanometer dimensions are significantly different from those of atoms and bulks materials. This is mainly due to the nanometer size of the materials which render them: (i) large fraction of surface atoms; (ii) high surface energy; (iii) spatial confinement; (iv) reduced imperfections; which do not exist in the corresponding bulk materials. These materials have created a high interest in recent years by virtue of their unusual mechanical, electrical, optical and magnetic properties. Nanophase ceramics are of particular interest because they are more ductile at elevated temperatures as compared to the coarse-grained ceramics. Nanostructured semiconductors are known to show various non - linear optical properties. Semiconductor Q-particles also show quantum confinement effects which may lead to special properties, like the luminescence.

Applications:

1. Single nanosized magnetic particles are mono-domains. Magnetic nano-composites have been used for mechanical force transfer (ferrofluids), for high density information storage and magnetic refrigeration.

2. Nanostructured metal-oxide thin films are receiving a growing attention for the realisation of gas sensors (NO_x, CO, CO_2, CH_4 and aromatic hydrocarbons).

3. Nanostructured semiconductors are used as window layers in solar cells.
4. Nano sized metallic powders have been used for the production of gas tight materials dense parts and porous coatings.
5. Carbon nanotube based transistors are used for miniaturizing electronic devices.
6. Carbon nanotube are used for making paper batteries.
7. A mixture of carbon nanotubes and fullerenes is used for making solar cells.
8. Microcrystalline TiO_2 is an insulator, whereas nano-crystalline TiO_2 is a semiconductor. Its Band gap can be tuned by controlling the size of nano-particle. Nano TiO_2 can be use for making dye sensitised solar cells.
9. Nanoparticles are used in the fight against tumours, nanostructured and functionalised surfaces and membranes improved diagnosis and more targeted use of active agents; neuro active implants.
10. Nano Cadmium telluride exhibit different colour depending upon its size. It can be used for dyeing fabrics, such nano colourants never fades.

2.1.3 Graphene

Graphene is an allotrope of carbon consisting of mono layer of carbon atoms arranged in a two-dimensional honey comb lattice (fig 2.1). It can be visualized as a single layer extracted from the layered structure of graphite. Graphite is three-dimensional whereas graphene is two dimensional with one atom thickness. Graphene is defined as "the two-dimensional mono-layer of carbon atoms, which is the basic building block of graphitic materials (i.e. fullerene, nanotube, graphite)".

It is considered to be one of the most promising materials of the century and gained world-wide attention due to its extraordinary charge transport, thermal, optical, and mechanical properties. It is considered to the lightest, thinnest, strongest material that conducts heat and electricity. It is stronger than diamond and ten times more conducting than copper.

Nobel Prize in Physics 2010 was awarded to Andre Geim and Konstantin Novoselov for the ground breaking experiments regarding graphene. There are various methods available for the synthesis of graphene, which includes the following.

1. Mechanical exfoliation method (such as Scotch tape method): Encompasses the repeated peeling off layers of graphite using adhesive tape. This was the method employed by Geim and Novoselov.
2. Chemical vapor deposition: Involves the deposition of carbon atoms onto a substrate (like copper) in the presence of a carbon-containing precursor gas such as methane.
3. Thermal decomposition on SiC: Silicon carbide substrate is heated under ultra-high vacuum which results in the sublimation of silicon atoms and deposition of carbon atoms to form graphene layers on the SiC surface.
4. Graphene oxide reduction method: Graphite oxide, obtained from graphite, is chemically treated to exfoliate into graphene oxide (GO), which is then reduced to graphene.

Different techniques such as optical microscope, atomic force microscopy (AFM), electron microscopy (SEM & TEM) are used for the characterization of graphene.

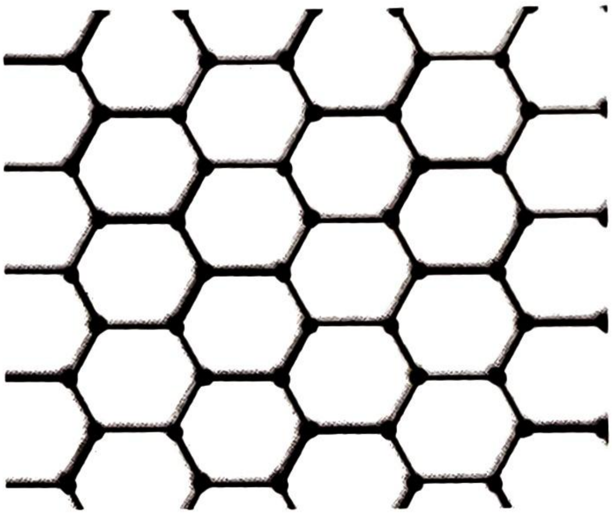


Figure 2.1: Honey comb lattice of graphene

In graphene the carbon atoms are sp^2 hybridized and are arranged in hexagonal fashion. Each hexagonal ring comprises of three strong in-plane sigma bonds, with a bond length of 0.142 nm, and a Pz orbitals perpendicular to the plane. These orbitals hybridize together to form two half-filled bands of free-moving electrons, π and π^*, which are responsible for most of graphene's notable electronic properties.

2.1.3.1 Properties of graphene

Electrical conductivity: Graphene has high electrical conductivity. Graphene, being two-dimensional material shows Quantum Hall effect. They behave as massless relativistic particles (Dirac fermions) which allows the electron speed comparable to light. They have high electron mobility compared to metals.

Mechanical strength and elasticity: They possess high elastic modulus and strength. It is 200 times stronger than steel. Graphene is highly is flexible and can be stretched by up to 20% of its original length without undergoing structural damage.

Thermal Conductivity: Graphene exhibits excellent thermal conductivity and is highly efficient for heat dissipation and thermal management applications.

Optical: Graphene absorbs only 2.3% of incident light over a broad wavelength range and hence makes it suitable for applications in transparent electrodes for displays, solar cells, touch-screens etc.

Surface area: It has high surface area due to its single-atom thickness and hence useful for applications in energy storage devices.

Graphene is chemically inert, stable and biocompatible also.

2.1.3.2 Applications of graphene

1. **Electronics:**

   (a) Due to its lower resistance and higher transparency, graphene-based thin film can be used in touchscreens, which is found to be superior than indium tin oxide.

   (b) Smaller size transistors can be developed using graphene which shows better performance.

2. **Energy storage:**

   (a) Graphene incorporated lithium-ion batteries have longer life span, faster charging time and higher capacity.

   (b) It can improve the efficiency of hydrogen fuel cell by lowering fuel cross over (fuel permeating through the electrolyte or membrane to the opposite side of the fuel cell).

   (c) Graphene is used in supercapacitors to provide high energy density.

3. **Biomedical:**

   (a) Suitably functionalized graphene can be used to carry chemotherapy drugs to cancer cells.

   (b) Graphene-based biosensors are highly sensitive when detecting DNA, ATP, dopamine etc.

4. **Composites and coating:**

   (a) By combining graphene with paint, a unique graphene coating is formed which will effectively prevent rusting.

   (b) Graphene in the carbon-fibre coating of aircraft's wing resists impact better and consumes less fuel.

   (c) Graphene-based composites and coatings could play a significant role in improving sports equipment for skiing, cycling etc.

5. **Environmental:**

   (a) Graphene based membranes can purify water in a more efficient, cheaper and environmental friendly way.

   (b) It can be used in filters and coatings to remove pollutants and toxins from the air.

2.1.4 Carbon Nanotubes

Carbon nanotubes (CNTs) are allotropes of carbon with cylindrical nanostructures. They are conceptually graphene sheet rolled into a tube. CNTs were discovered in 1991 by the Japanese electron microscopist Sumio Iijima who was studying the material deposited on cathode during the arc-evaporation synthesis of fullerenes. There are different methods available for the synthesis of CNTs with different structure and morphology. Arc-discharge method, laser ablation/evaporation method, chemical vapor deposition method etc. are the commonly used methods for the synthesis of CNTs.

Carbon nanotubes are one-dimensional nano particles with diameter measuring 1–50 nm and are generally only a few micrometres in length. They are molecular scale tubes and are nanoscopic hollow fibres of pure carbon. It is 10^5 times thinner than human hair. They are characterised by high tensile strength and according to structure are conductive or semi conductive. CNTs are at least 100 times stronger than steel, but only one-sixth as dense. In addition, they conduct heat and electricity far better than copper.

2.1.4.1 Classification of CNTs

There are two types of CNTs namely single-walled (SWCNT, one tube) and multi-walled (MWCNT, several concentric tubes). Both of these are typically a few nanometres in diameter and several micrometres to centimetres long.

SWCNT can be visualized as rolled-up tubular shell of graphene sheet. Based on the variations arising from the specific orientation and rolling of graphene sheets, SWCNTs are classified into three viz, armchair, zigzag, and chiral. Armchair nanotubes exhibit a symmetrical arrangement where the rolling axis aligns with the hexagonal lattice of graphene, resulting in equal edge lengths. Zigzag nanotubes, on the other hand, are formed when the rolling axis is along the zigzag pattern of the graphene lattice, leading to edges that consist solely of zigzag lines. Chiral nanotubes are characterized by a rolling axis that is oriented at an angle to the graphene lattice, resulting in unequal edge lengths and a helical structure. The optoelectronic properties of carbon nanotubes vary significantly with molecular structure and diameter of the tube.

MWCNT is a stack of graphene sheets rolled up into concentric cylinders. There are three different models proposed for MWCNT namely Russian doll model, Parchment model and mixed model. In Russian doll model, sheets of graphene are arranged into concentric cylinders whereas in Parchment model single graphene sheet is rolled around itself, similar to a scroll of parchment. Mixed model is a mixture of both Russian doll and Parchment model. Classification of CNTs is shown in figure 2.2.

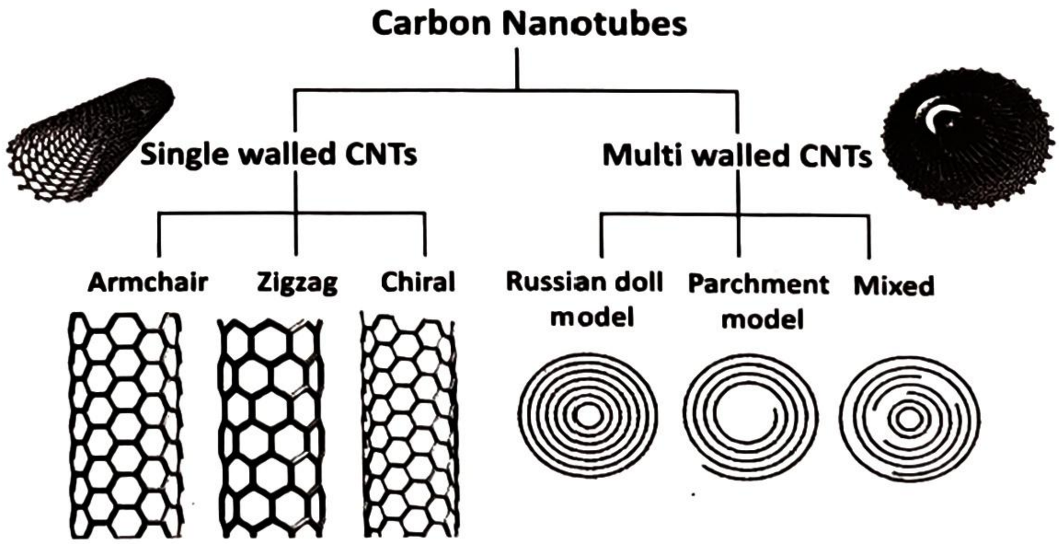


Figure 2.2: Classification of carbon nanotubes

2.1.4.2 Properties of Carbon Nanotubes:

Strength and hardness: They are the strongest and stiffest materials. Single walled nanotubes (SWNTs) are used for synthesizing super-hard material, by compressing it at room temperature. High strength could be attributed to the covalent sp^2 bonds formed between the individual carbon atoms. They can withstand a pressure upto 24 GPa without deformation. Multi walled nanotubes (MWNT) without inter connected inner shells exhibit telescoping property.

Electrical conductivity: Their electrical conductivity is better than metals. Electron travelling through a CNT behaves like a wave travelling through smooth channel- Ballistic Transport. MWNTs with interconnected inner shells show superconductivity with relatively high temperatures.

Thermal conductivity: Their thermal conductivity also is better than metals. All nanotubes are expected to be very good thermal conductors along the tube but good insulators laterally to the tube axis- Ballistic Conduction. The thermal stability of carbon nanotubes is found to be upto 2800^0C in vacuum and about 750^0C in air.

2.1.4.3 Application of Carbon Nanotubes:

1. **Electrical circuits:** Nanotube based transistors have been made that operates at room temperature. Carbon nanotubes are used for miniaturizing electronic devices.
2. **Energy storage:** Due to high surface area, optimised electrical and thermal properties they are widely employed in energy storage applications.
   (a) Paper Batteries: A paper battery is a battery engineered to use a paper-thin sheet of cellulose infused with aligned carbon nanotubes. It gives a steady power output. This battery also functions as a super-capacitor which give a quick explode of high energy.
   (b) Solar Cells: CNT-fullerene hybrid solar cell is formed by a mixture of carbon nanotubes and fullerenes. Electrons trapped inside fullerenes are excited by sunlight leading to the flow of this electrons, which produce the current. CNT acts as the conductive pathway.
   (c) They are used for making ultra-capacitors which provide a large surface to store electrical charge
3. **Aerospace and automotive industry:** Due to easier molding, high strength, stiffness, and reduced weight, CNT polymer composites are preferred for aerospace and automotive applications.
4. Research is also being carried out to assess the ability of CNTs for the storage of hydrogen.

2.1.5 Carbon Quantum Dots (CQDs)

Carbon quantum dots are zero-dimensional small carbon nano particle. It consists of ultrafine, distributed, quasi-spherical carbon nanoparticles with size less than 10 nm. Recently they have attained much attention due to their good solubility and strong luminescence properties. CQDs are found to be much more superior than traditional semiconductor quantum dots due to its lower molecular weight, reduced toxicity, ease of surface functionalization, cost-effectiveness, exceptional fluorescence stability, tunable emission wavelengths, strong biocompatibility etc.

Carbon quantum dot was first discovered by Xu et al. in 2004 accidentally during the purification of single-walled carbon nanotubes via electrophoresis. It can be synthesised using two main methods: 'top-down' and 'bottom-up'. The top-down approach involves breaking down large carbon structures into small CQDs using techniques like chemical oxidation, laser ablation, arc discharge, and electrochemical synthesis. The bottom-up method involves building CQDs from small carbon molecules through processes like hydrothermal synthesis, microwave-assisted synthesis, and pyrolysis, which allows control over their size and shape. Synthesised particles can be purified using electrophoresis, centrifugation, dialysis, column chromatography etc.

2.1.5.1 Properties

Optical: CQDs shows absorption mainly in the UV region which can extent to the visible region also. They exhibit strong fluorescence, photoluminescence and chemiluminescence which makes them useful in imaging and sensing applications. They can be excited by various wavelengths of light and often display tunable emission spectra.

Stability and biocompatibility: CQDs show excellent photostability and biocompatible with less toxicity than other quantum dots and hence suitable for medicinal and biological applications.

2.1.5.2 Applications

1. **Biomedical:**
    (a) Due to its fluorescence property, better biocompatibility and low biotoxicity they are employed in fluorescent bioimaging.
    (b) CQDs are used as biosensor carriers because of their solubility in water, tunable excitation property, excellent biocompatibility and photostability. CQDs-based biosensors can visually monitor various substances including glucose, copper, phosphate, iron, potassium, and nucleic acids.
    (c) They are used in drug delivery systems also.
2. **Optoelectronics:**
    (a) CQDs have shown potential in enhancing the performance of DSCs due to their stable light absorption properties, photostability and low cost. By creating CQD-bridged dye/semiconductor complex systems, CQDs improve the photoelectric conversion efficiency significantly.
    (b) CQD-based hybrids have been identified as excellent materials for supercapacitors. CQD-RuO_2 hybrid show remarkable electrochemical performance.
3. **Lighting:** Due to their stable light emitting, low cost and eco-friendliness they are used as LED materials. Switchable electroluminescence behaviour makes it useful for developing colourful LEDs.

4. **Environmental:** CQDs can be used to improve the photocatalytic degradation of organic pollutants and dyes in water. Their ability to absorb light and facilitate charge separation enhances the efficiency of photocatalysts like TiO_2 in breaking down harmful organic compounds.

5. **Catalysis:** CQDs-modified P25 TiO_2 composites (CQDs/P25) enhance photocatalytic hydrogen evolution.

2.1.6 Fullerenes

Fullerenes are zero-dimensional, hollow, closed cage nanoparticles made of carbon atoms. The discovery of fullerenes in 1985 by Curl, Kroto, and Smalley culminated in their Nobel Prize in 1996. Fullerenes, or Buckminster fullerene (fig. 2.3), are named after Richard Buckminster Fuller the architect and designer of the geodesic dome and are sometimes called bucky balls.

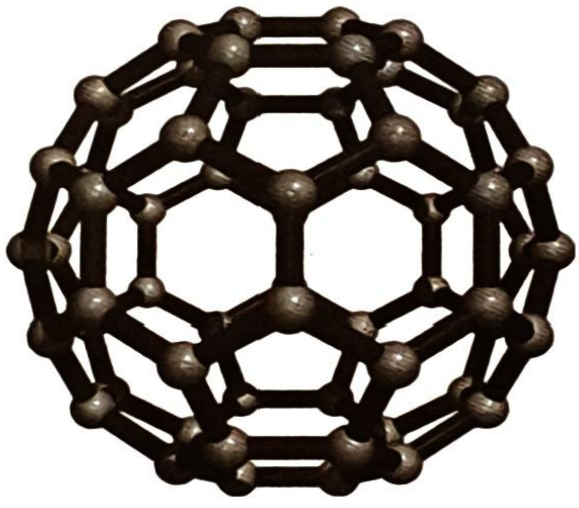


Figure 2.3: C60 "bucky ball" fullerene

Fullerene molecules are denoted based on the number of carbon atoms in the spherical carbon ball. The smallest fullerene was reported with twenty carbon atoms as C20 followed by C24 and C28 analogues. Each fullerene generally contains 12 pentagonal and (n/2-10) hexagonal rings, where n $\geq$ 20.

C60 fullerene, known as the Buckminster fullerene is the primarily discovered and most widely studied form of fullerene. In 1990, a technique to produce larger quantities of C60 was developed by heating graphite rods in a helium atmosphere. The sooty material formed by condensation of vaporized carbon consists of mainly C60 with smaller quantity of C70 (rugby ball shape) sand traces of fullerenes consisting of even number of carbon atoms up to 350 or above. It is composed of fused pentagonal and hexagonal carbon rings. It contains 12 five membered rings ($12 \times 5 = 60$ atoms) and 20 six membered rings, possess a perfect icosahedral symmetry. The geometry is same as that of soccer football.

Fullerenes are stable, but not totally unreactive. The sp^2 hybridized carbon atoms, which are at their energy minimum in planar graphite, must be bent to form the closed sphere or tube, which produces angle strain. The characteristic reaction of fullerenes is electrophilic addition at 6,6-double bonds, which reduces angle strain by changing sp^2 hybridized carbons into sp^3 hybridized ones. The change in hybridized orbitals causes the bond angles to decrease from about 120^0 in the sp^2 orbitals to about 109.5^0 in the sp^3 orbitals. This decrease in bond angles allows for the bonds to bend less when closing the sphere or tube, and thus, the molecule becomes more stable.

2.1.6.1 Properties

Physical: Fullerenes are extremely strong molecules, able to resist great pressures - they will bounce back to their original shape after being subject to over 3,000 atmospheres. This property makes fullerenes become harder than steel and diamond. An interesting experiment shows that Fullerenes can withstand collisions of up to 15,000 mph against stainless steel, merely bouncing back and keeping their shapes. This experiment demonstrates the high stability of the molecule.

Solubility: Fullerenes are sparingly soluble in many solvents. Common solvents for the fullerenes include aromatics, such as toluene, and others like carbon disulfide. Solutions of pure buckminster fullerene have a deep purple color. Solutions of C70 are a reddish brown. The higher fullerenes C76 to C84 have a variety of colors.

Electrical: Fullerenes are normally electrical insulators, but when crystallized with alkali metals, the resultant compound can be conducting or even superconducting.

Light absorption: Fullerenes absorb strongly in the UV and moderately in the visible regions of the spectrum.

Chemical: The carbon atoms within a Fullerene molecule are sp^2 and sp^3 hybridized, of which the sp^2 carbons are responsible for the considerably angle strain presented within the molecule. C60 and C70 exhibit the capacity to be reversibly reduced with up to six electrons.

2.1.6.2 Applications

1. **Coatings and lubricant:**
   (a) Fullerenes are used for making durable, high-performance coatings that resist wear and corrosion.
   (b) Fullerenes are used as miniature 'ball bearings' to lubricate surfaces. C60 can be used used as excellent microscopic ball bearings, lubricant and catalyst.
2. **Biomedical:**
   (a) Fullerenes can be used to deliver drugs to specific cells or tissues due to their ability to form stable complexes with various molecules.
   (b) Fullerenes can be used as contrast agents in imaging techniques such as MRI or ultrasound.
3. **Energy storage:**

(a) Fullerenes and their derivatives are employed energy storage devices due to their electrical conductivity and stability.

(b) They can be used to store hydrogen.

4. **Environmental:**

(a) Used in water treatment for the removal of pollutants and contaminants.

(b) Used in sensors for detecting environmental toxins and pollutants.

2.2 Polymers

Polymers are molecules with large molecular masses obtained by the covalent linkage of several small repeating chemical units called monomers. The repeating chemical unit may be same or different. If there is only one type of monomer in a polymer then it is known as homopolymer, eg. Polyethylene, Poly propylene, PVC, etc. The number of repeating units is called the degree of polymerization. A co-polymer is made of two or more different monomeric species, eg. butadiene-styrene (BS) Acrylonitrile butadiene-styrene (ABS), etc. Based on the types of polymerizations, polymers can be broadly divided into addition (eg; PVC, neoprene etc.) and condensation polymers (PET, bakelite etc). Two special classes of polymers viz., fire-retardant polymers and conducting polymers will be discussed in this chapter.

2.2.1 Fire-retardant Polymers

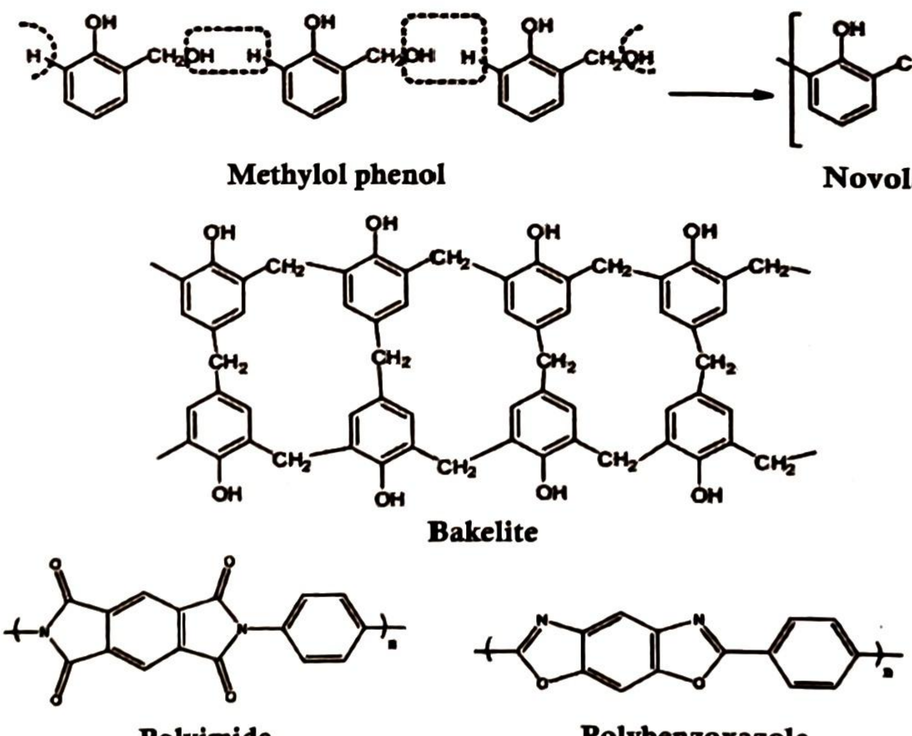


Figure 2.4: Examples for non-halogenated fire-retardant polymers.

Most of the polymers contain organic compounds as monomers and are flammable. Fire-retardant polymers resist ignition and slow the spread of flames. Nitrogen based polymers such as melamine-formaldehyde, inorganic polymers such as silicone polymers etc comes under the category of fire-retardant polymers. Phenolic polymers such as novolac, bakelite etc and aromatic heterocyclic polyimides, polybenzoxazoles etc are also examples of intrinsically fire-resistant polymers (fig 2.4).
Inorganic additives such as aluminium hydroxide, magnesium hydroxide, ammonium polyphosphate etc. and organophosphorus compounds (such as DMMP, BDP) can be incorporated into polymers to enhance their fire resistance.

Halogenated polymers are also fire-retardant polymers. Fire is a free radical chain reaction system which contains radicals such as $H^\bullet$, $OH^\bullet$, $OOH^\bullet$, $CH_3^\bullet$ etc. In the case of halogenated polymers like PVC, Poly brominated styrene etc halogen radical ($Cl^\bullet$, $Br^\bullet$) is also introduced into the fire, which combines with H radical and OH radicals to form stable molecules like HX and HOX (where X can be Cl or Br) respectively, resulting in quenching of chain reaction. Among halogens Bromine is found to be more effective in quenching fire.

Examples of chlorine containing polymers are polyvinylchloride (PVC) and chloroprene rubber (neoprene). Poly(bromo styrene) and brominated butyl rubber are examples of bromine containing polymers. Brominated butyl rubber is synthesised by the bromination of butyl rubber. Structures of the polymers are given in fig 2. 5.

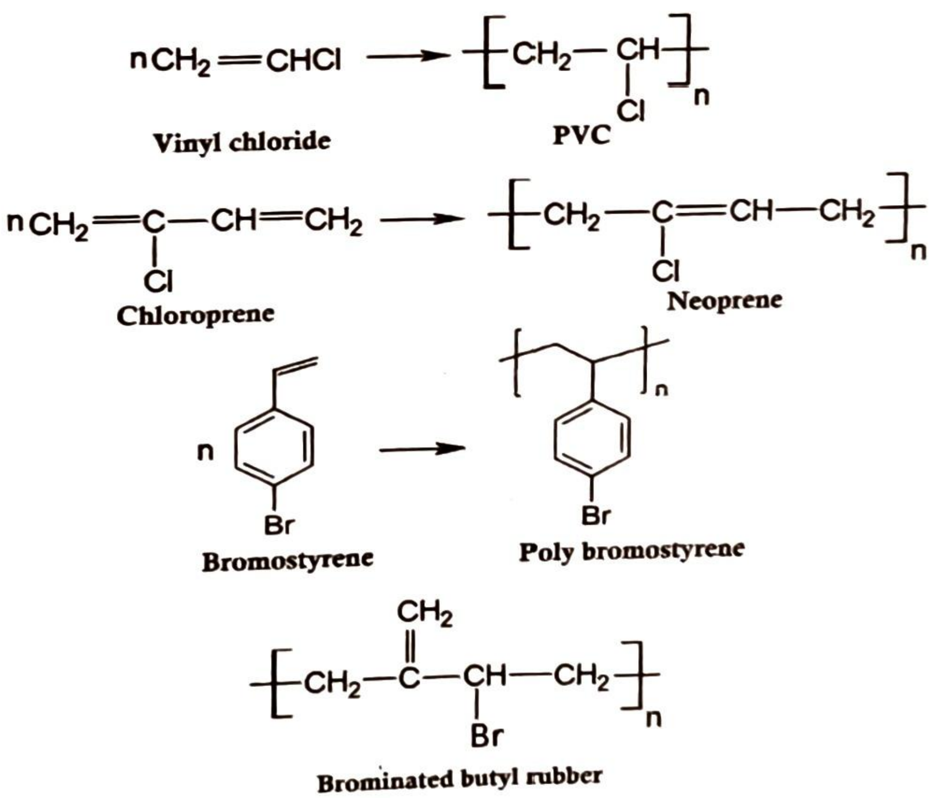


Figure 2.5: Examples for halogenated fire-retardant polymers.

Use of halogenated polymers are not much recommended since they remain in the environment for long periods and can release toxic by-products during combustion or degradation.

Note: Apart from these halogenated polymer, halogenated compounds, especially those containing bromine or chlorine, have been used as fire retardants. They will release halogen atoms and interfere with the combustion process.

Examples include polybrominated biphenols (PBBs), polybrominated diphenyl ethers (PBDEs), tetrabromobisphenol A (TBBPA) and hexabromocyclododecane (HBCD), polychorinated biphenyls (PCBs) and trichloroethylene (TCE). Structure of the compounds are given in fig 2.6.

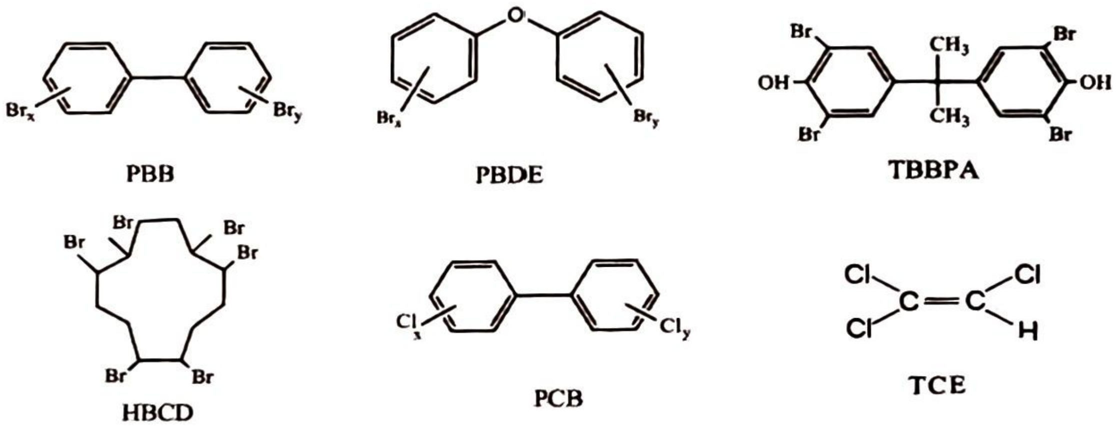


Figure 2.6: Halogenated fire-retatrdants

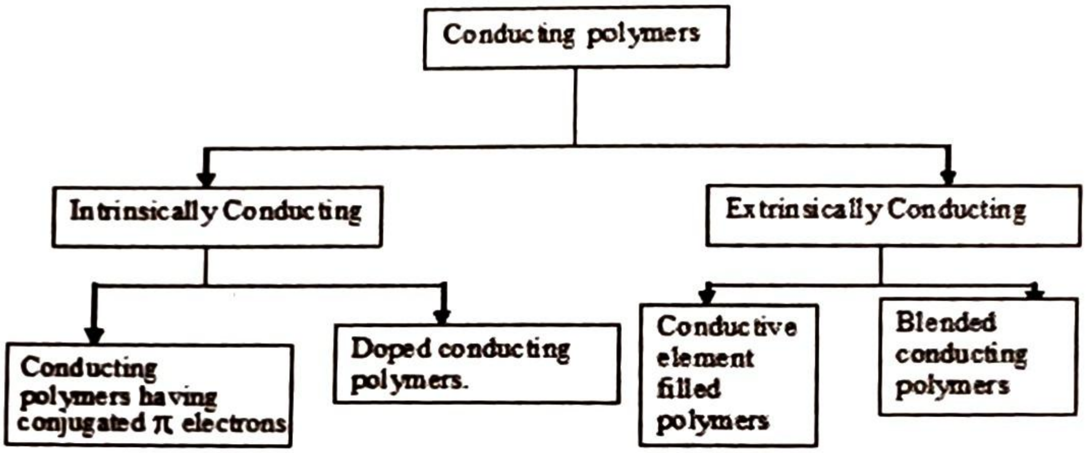
2.2.2 Conducting Polymers

In general, polymers are found to be insulators. With in the past few years, a new class of organic polymers have been synthesized which possess remarkable ability to conduct electrical current. A polymer which can conduct electricity is termed as conducting polymer. Among many polymers known to be conductive, poly acetylene, poly aniline and poly pyrrole have been studied most. However, the conductive polymer that was actually launched this new field was poly acetylene.

Classification: Conducting polymers can be classified into following types

2.2.2.1 Intrinsically conducting polymers

These polymers have extensive conjugation in the backbone which is responsible for conductance. This is again classified as



1. Conjugated π - electron conducting Polymers:

A key property of a conducting polymer is the presence of conjugated double bonds along the backbone of the polymer. The orbitals of conjugated π - electrons overlap over the entire backbone of the polymer, resulting in the formation of valence bands and conduction bands, which extends over the entire polymer molecule. The highest occupied band is called valence band and lowest unoccupied band is the conduction band. These bands are separated by significant band gap. Thus electrical conduction would occur when electron from valence band are excited to conduction band either thermally or photolytically eg. polyacetylene and polyaniline.

acetylene polyacetylene

aniline poly aniline

2. **Doped conducting polymers:**

In comparison to conventional polymers the conducting polymers with conjugated π-electrons in their backbone can be easily oxidized or reduced as they have low ionization potentials and high electron affinities, ie; their conductivities can be increased by creating positive or negative charge on polymer backbone by oxidation or reduction. This process is referred to as 'Doping' and are of two types

(a) **p-doping:** It is done by oxidation process. In this process some electrons from π - bond of the conjugated double bonds are removed and the holes so created can move along the molecule, ie; the polymer becomes electrically conductive. The radical cation produced is called 'polaron'. The polarons are mobile and can move along the polymer chain by rearrangement of double and single bonds and hence the polymer become conducting. The oxidation process is generally brought about by Lewis acids such as $FeCl_3$.

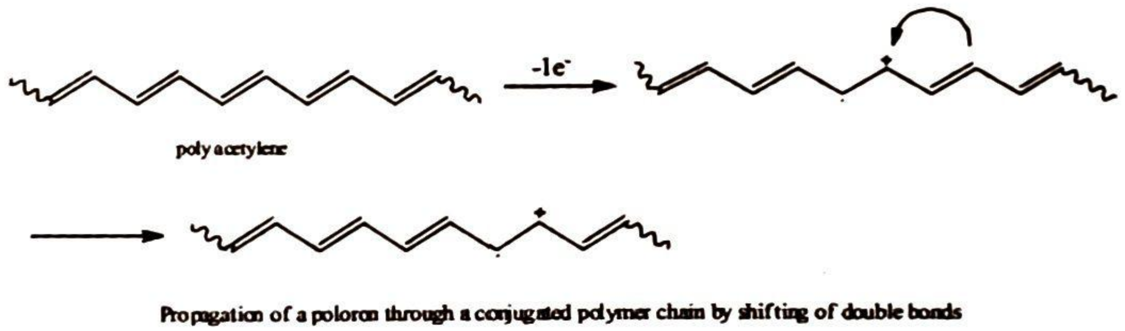


Figure 2.7: p-doping (polaron movement)

(b) **n-doping:** In this type of doping some electrons are introduced into the polymer having conjugated double bonds by reduction with Lewis bases like lithium, sodium naphthalide etc. The reduction of polyacetylene by a Lewis base leads to the formation of anion called 'polaran' and bipolaran in two steps.

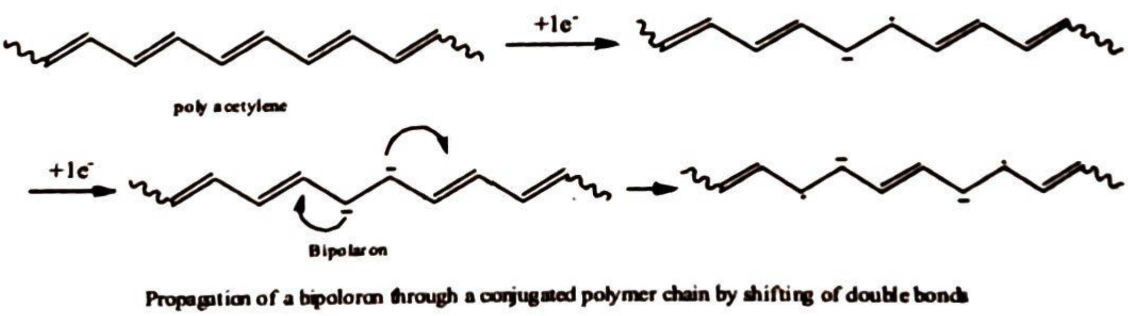
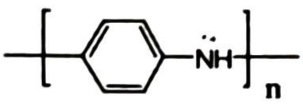
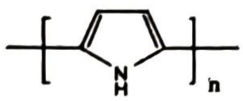


Figure 2.8: n-doping (polaran movement)

Conductive polymers	Repeating unit	Dopent		Scm⁻¹
		n type	p-type	
Trans-poly acetylene		Li, AsF_5	I_2, Br_2	10^4
Poly aniline			HCl	200
Poly pyrrole			BF_3 $HClO_4$	500-7500

2.2.2.2 Extrinsically conducting polymers

There are conducting polymers whose conductivity is due to the presence of externally added ingredients in them.

1. **Conductive element filled polymer:** These are polymers which are filled with conducting elements such as carbon black, metallic fibers, metal powders etc. The polymer acts as a binder to hold the conducting elements together. These polymers are, (i) low in cost, (ii) light in weight (iii) mechanically durable and strong, (iv) easily processable in different forms, shapes and sizes. This type is used in making computer key boards.

2. **Blended conducting polymers:** These polymers are obtained by blending conducting polymers with conventional polymers. These polymers possess better physical, chemical and mechanical properties.

2.2.3 Polyaniline (PANI)

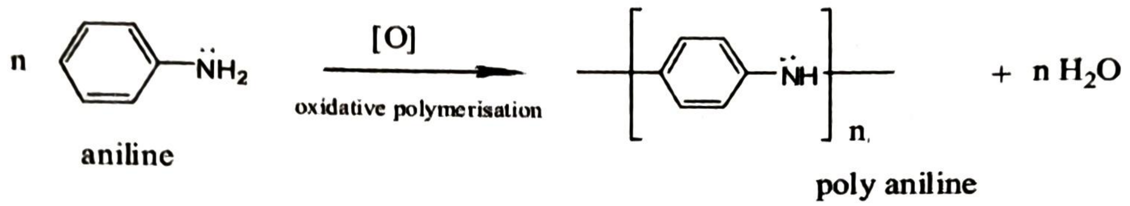
The oxidative product of aniline under acidic conditions is known as aniline black from 1862. Since 1980s polyaniline captured the intense attention of the scientific community due to the rediscovery of high electrical conductivity.

Chemical synthesis

1. **Chemical method:** Aqueous solution of ammonium per sulphate is added slowly to a solution of aniline dissolved in dilute hydrochloric acid at a temperature of 5^0C under agitation. After an hour a precipitate is formed, which can be removed by filtration. The polymer thus obtained is washed with ammonium hydroxide solution and dried.

2. **Electro-chemical method:** When thin film or better ordered polymer is required an electrochemical oxidation method is preferred. It is done by dipping platinum foil electrodes on acidified aqueous solution of aniline and a potential of 0.7 V is applied, then polyaniline film is formed on the surface of the positive electrode.

Properties Among various conducting polymers, polyaniline (PANI) is a unique and promising candidate for potential applications because of its stability in air and solubility in some



organic solvents. It also exhibits dramatic changes in its electronic structure and physical properties on protonation. Recently PANI has attracted much attention for its magnetic behaviour as it exhibits high spin density and ferromagnetic spin-spin interaction.

The colour change associated with polyaniline in different oxidation states can be used in sensors and electrochromic devices. Polyaniline is more noble than copper and slightly less noble than silver which is the basis for its broad use in printed circuit board manufacturing and in corrosion protection. Doped poly aniline is a good conductor comparable to copper, whereas the original form is a semiconductor.

Applications

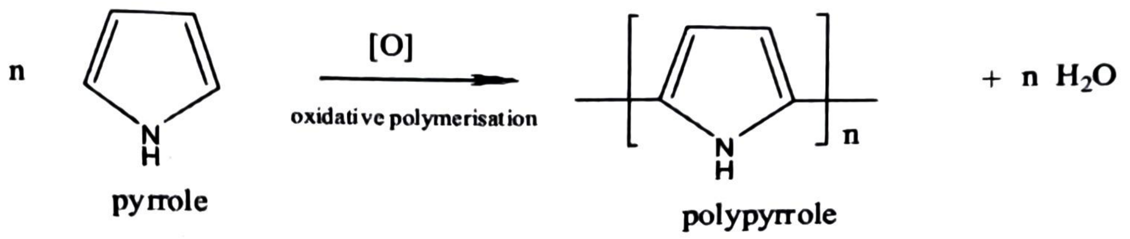
1. As electrode material for rechargeable batteries.
2. In the dissipation of static electric charges in metallic containers.
3. As shield in complex electronic circuits and electromagnetic radiations.
4. In biosensors and solar energy absorption.

2.2.4 Polypyrolle

Polypyrrole is one of the most promising material with multifunctionalised applications. It is an intrinsic conducting polymer with conjugated double bonds.

Preparation: It can be prepared by electrochemical or chemical polymerization.

1. **Electro-chemical method:** Polypyrrole can be synthesized by electrochemical technique as follows. By the electrolysis of pyrrole (0.06 M) in aqueous acetonitrile solution using platinum electrodes and tetraethyl ammonium boron tetrafluoride as the supporting electrolyte (0.01 M), the polymer will be precipitated as a blue black film on the anode.
2. **Chemical method:** Polypyrrole can be synthesized by chemical oxidative polymerization technique using monomer pyrrole and ammonium per sulfate as oxidant in the ratio 1:1. The chemical polymerization can be carried out in a beaker by mixing 0.1 M aqueous solution of pyrrole and 0.1 M of ammonium per sulfate in 1:1 ratio for 3 hrs. The polymer precipitates out. The PPy films can also be deposited on pre-cleaned glass substrate.



Properties: PPy has excellent electrical, thermal and mechanical properties. It has good environmental stability and higher conductivity. The electrical resistance was found to decrease as temperature increases and hence conductivity increases. The decrease of resistance is due to the increase of efficiency of charge transfer in PPy with increase in temperature. PPy films are yellow but darken in air due to oxidation. Doped films are blue black depending on the degree of polmerisation and film thickness. They are amorphous and give weak diffraction. Pure PPy is an insulator while its oxidative derivatives are good electrical conductors.

Applications of Polypyrrole (PPy)

1. Polypyrrole is used mostly in biosensors and immunosensors, because of its biocompatibility.

2. Compared with metal and inorganic materials, doped PPy has better mechanical match with live tissue, resulting in its many applications in biomedical field.

3. It have applications in tissue engineering.

4. It is also employed as counter electrodes in dye-sensitized solar cells.

2.2.5 Applications of conducting polymers

1. In Rechargeable Batteries: The conducting polymers can be used as cathodes and solid electrolytes in batteries for automotive and other applications. These batteries have high reliability, light weight, non leakage of electrolyte and are small in size.

2. Some conducting polymers like polyaniline show different colours in different oxidation forms. This electrochromic property can be used to produce "smart windows" and electrochromic displays. [Smart windows are the windows which change colour in response to sunlight or temperature changes].

3. These are also used in electro-luminascence OLED displays like TV, mobile phone etc.

4. In analytical sensors: These polymers are used for making sensors for pH, O_2, NO_x, SO_2 and glucose detection.

5. In photovoltaic devices: eg. In Al, polymer/Au photovoltaic cells.

6. In electronics: It can be used in Light Emitting Diodes (LEDs), electromagnetic shielding, electron beam lithography, etc.

2.3 Organic Light Emitting Diodes (OLEDs)

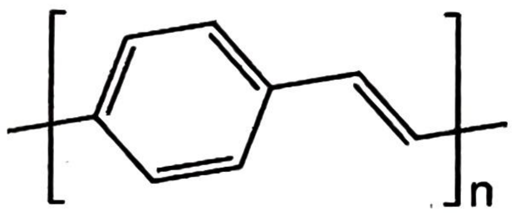
An organic light - emitting diode (OLED) is a light-emitting diode (LED) in which the emissive electroluminescent layer is a film of organic compound which emits light in response to an electric current. This layer of organic semiconductor is situated between two electrodes; at least one of these electrodes is transparent. There are two main families of OLED: (i) those employing conducting polymers (ii) those based on small molecule.

Principle: The principle of conjugated polymer OLED is HOMO-LUMO transitions. The basic principle of small molecule OLED is the creation of a light-emitting electrochemical cell (LEC) on adding mobile ions to an OLED.

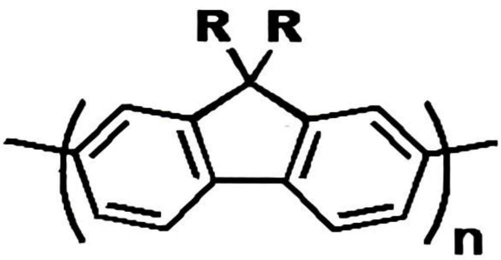
OLED employing conducting polymers

A typical OLED is composed of a layer of organic materials situated between two electrodes, the anode and cathode, all deposited on a substrate. The organic molecules are electrically conductive as a result of delocalization of π-electrons caused by conjugation of the molecule. These materials have conductivity levels ranging from insulators to conductors, and are therefore considered organic semiconductors. The highest occupied and lowest unoccupied molecular orbitals (HOMO and LUMO) of organic semiconductors are analogous to the valence and conduction bands of inorganic semiconductors like silicon.

The organic compounds used in OLEDs include derivatives of poly(p-phenylene vinylene) (PPV) and polyfluorene. Substitution of side chains onto the polymer backbone may determine the colour of emitted light.



Poly(para-phenylene vinylene



polyfluorene

Construction: Many modern OLEDs incorporate a simple bilayer structure, consisting of a conductive layer and an emissive layer. A typical organic light emitting diode (OLED) structure consisting of a glass substrate, a indium tin oxide (ITO) anode, and an organic bilayer diffused into a metallic cathode, as given below. The organic bilayer consists of a hole transport layer (HTL) and an emitting/electron transport layer (EML/ETL). The emissive layer and the conducting layer is made up of organic molecules (both being different). The emissive layer used in an OLED is made up of n-type polymer molecules, out of which the most commonly used is polyfluorene. The conducting layer is made up of p-type polymer, and the commonly used component is polyaniline.

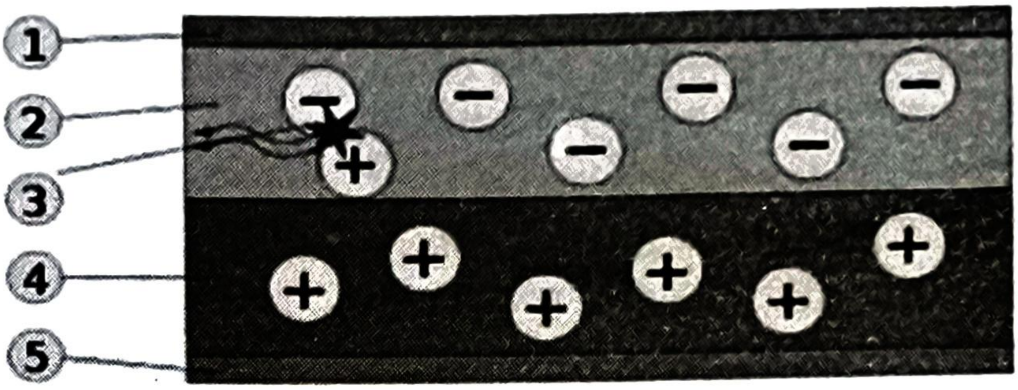


Figure 2.9: Schematic of a bilayer OLED 1. ITO anode(-), 2. Emissive layer, 3. Emission of radiation, 4. Conductive layer, 5. Metal cathode (+)

Working: To make an OLED light up, we simply attach a voltage (potential difference) across the anode and cathode. As the electricity starts to flow, the cathode receives electrons from the power source and the anode loses them. Now we have a situation where the added electrons are making the n-type emissive layer negatively charged, while the p-type conductive layer is becoming positively charged. Positive holes are much more mobile than negative electrons so they jump across the boundary from the conductive layer to the emissive layer. When a hole (a single electron π-bond) meets an electron, the two things cancel out and release a brief burst of energy in the form of a quantum of light or a photon, in other words. This process is called recombination, and because it's happening many times a second the OLED produces continuous light for as long as the current keeps flowing.

Note: In conventional LED like GaAs and GaN electrons are more mobile than holes so p-region is the emissive layer.

Small-molecule OLED

It consists of three organic layers sandwiched between electrodes. The organic layers adjacent to cathode and anode are the electron transport layer (ETL) and the hole transport layer (HTL), respectively. Emissive layer (EML) usually consists of light-emitting dyes or dopants dispersed in a suitable host material. Molecules like quinacridone derivatives are often used. Aluminium quinacridone (Alq3) has been used as a green emitter (fig.2.10).

Advantages of OLED

1. The manufacture of OLED is highly economical and is more efficient than LCD and flat panel screens.
2. OLED display can be printed cloths and fabrics by ink jet printing. This technology will help in carrying huge displays in our hands.
3. There is much difference in watching a high-definition TV to a OLED display. As the contrast ratio of OLED is very high (even in dark conditions), it can be watched from an angle nearly 90 degrees without any difficulty.

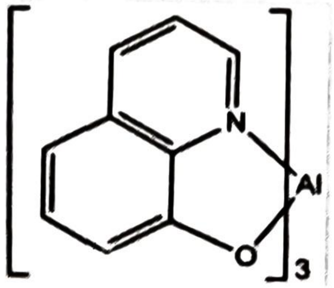


Figure 2.10: Aluminium quinacridone (Alq3)

4. No backlight is produced by this device and the power consumption is also very less.
5. OLED has a refresh rate of 100,000 Hz which is almost 9900 Hz greater than an LCD display.
6. The response time is less than 0.01 ms. LCD needs a response time of 1 ms.

2.4 Dye-Sensitized Solar Cells (DSSCs)

Escalation in global population and energy consumption demand the urgent shift towards renewable energy sources. Among various renewable energy sources, solar energy stands out as the most practical due to its efficiency, low fabrication cost, and environmental benefits. A solar cell is an electrical device that converts the energy of light directly into electricity by the photovoltaic effect. Conventional solar cells use crystalline silicon semiconductors. It works on the principle of photovoltaic cell. Here photons of light knock electrons from atoms within the semiconductor material, creating an electric current.

DSSCs were developed as an efficient and environmentally benign alternative to silicon based solar cells. A dye-sensitized solar cell (DSSCs) is a photoelectrochemical cell, which converts solar light absorbed by the dye to electric energy. It is a thin-film solar cell.

2.4.1 Construction of Dye-sensitized solar cells

A DSSC mainly consists of a substrate, transparent conductive oxide layer (TCO), semiconductor metal oxide nano particle, organic / inorganic dye, electrolyte and counter electrode.

Substrate is generally made up of glass or flexible transparent plastic, TCO is coated on the substrate.

Transparent conductive oxide (TCO): Thin film of Indium Tin Oxide (ITO) or Fluorine-doped Tin Oxide (FTO) acts as TCO.

Semiconductor metal oxide nano particles: Semiconducting metal oxides with high surface area and wide band gaps are used in DSSCs. TiO_2 is the commonly used metal oxide. Other metal oxide like ZnO and SnO_2 are also used.

Dye/photo sensitiser: Surface of the metal oxide particle is coated with dye molecules. These dye molecules are responsible for absorbing sunlight and inject the photoexcited electrons to the conduction band of the semiconductor. Dye-sensitized metal oxide semiconductor acts as the anode. Ruthenium-based complexes and organic dyes are commonly used. Natural dyes extracted from plants can also be used as photo sensitiser.

Electrolyte: Redox couple iodide/triiodide solution is used as the electrolyte. Electrolyte should have good contact with the dye-sensitized metal oxide semiconductor and the counter electrode.

Counter electrode: Platinum or graphite-based materials are employed as the counter electrodes, which acts as the cathode. It facilitates the reduction of the oxidized dye molecules and completes the electrical circuit.

Schematic representation of DSSC employing titanium dioxide semiconductor and iodide/triiodide electrolyte is given in fig. 2.11.

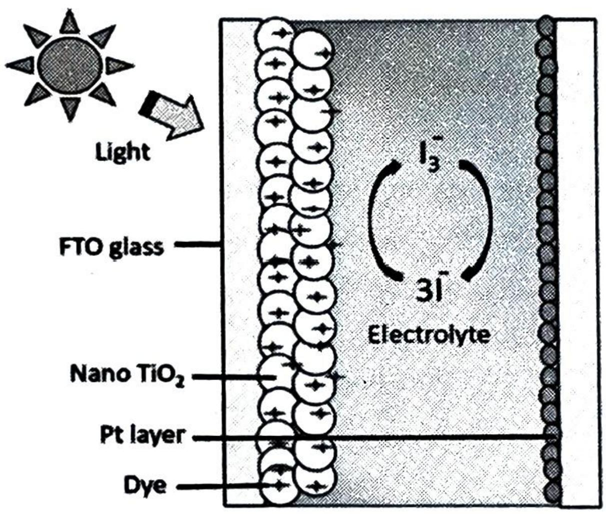


Figure 2.11: Schematic representation of DSSC

(Source: https://doi.org/10.1016/j.molstruc.2022.133512)

2.4.2 Working of Dye-sensitized solar cells

Sunlight passes through the transparent conductive oxide layer and the dye molecules coated on the metal oxide semiconductor absorb the light. The electrons of the dye molecule then get excited and are injected into the conduction band of the metal oxide semiconductor. These electrons travel through metal oxide and move towards the external circuit to generate electric current. Electrons reach the counter electrode through the circuit and generate electricity. The oxidized dye molecules will get reduced by gaining electrons from the electrolyte and come back to their original form. The oxidized electrolyte species diffuse towards the counter electrode and get reduced. This process continues to produce the current.

2.4.3 Advantages of DSSC

- The fabrication of DSSCs is simpler processes compared to conventional silicon solar cells.
- They are light weight which can be advantageous for applications on vehicles and portable devices.
- They are flexible in shape, and size and hence can be implemented on windows, terraces, facades, etc.
- Natural pigments and organic dyes can be employed makes them more sustainable, ecofriendly and affordable.

2.4.4 Applications of DSSC

DSSCs are mainly used for sunlight harvesting.

- Since they can be used in window, facades etc., they can generate power while maintaining aesthetic appearance.
- They can be used in portable solar chargers for electronic devices, such as smartphones and tablets due to its lightweight.
- They can be integrated into the fabric of bags and backpacks, enabling them to charge electronic devices while on the go. This feature makes them particularly popular among hikers and travellers who need a portable and convenient power source.
- DSSC can be used to power illuminated signs and displays in locations where access to electricity is limited or costly.

2.5 Spintronics

Spintronics or spin-electronics is an emerging field that exploits the spin of electrons, in addition to charge, to process information in new generation transistors, lasers and integrated magnetic sensors. They act by storing information in electron spins, transmitting this information via mobile electrons and reading it at a terminal. Spin orientation lasts longer than charge-based information, making them suitable for memory storage, magnetic sensors and quantum computing.

Spintronics is basically connected to magnetism since the magnetic moment of electron is proportional to its spin. Spintronics make use of the magnetoresistive effects for sensing and storing information. The discovery of giant magnetoresistance (GMR) has a crucial role in the development of spintronics. GMR is a phenomenon where a material's electrical resistance changes dramatically when exposed to a magnetic field. This effect is widely used in magnetic storage (eg. Magnetic random-access memory) and sensing technologies. Albert Fert and Peter Grünberg, were awarded the Nobel Prize in Physics in 2007 for this ground breaking discovery. Giant magnetoresistive spin-valve sensor have improved the capacity of hard disc drive by more than hundred-fold times. This sensor comprises a metallic stack of ultra-thin ferromagnetic layers separated by a thin copper layer.

A spin polarised current can be generated in a very simple way by passing the current through a ferroelectric material. Spintronics materials should retain ferromagnetism practical temperatures. There are metal based, semiconductor based and graphene based spintronics. Metal oxides such as MgO, Al_2O_3, $SrTiO_3$ etc are examples of metal based spintronics materials. Ternary alloys such as InAs and GaAs alloyed with Mn, (Co, Fe, B), (Ga, Fe)N etc are used as semiconductor based spintronic materials due to their tunable magnetic and electronic properties. Graphenes are also used for development of spintronics due its efficient spin manipulation properties.

2.6 Materials in Quantum Computing Technologies

Quantum computing is an emerging computer technology that uses the principle of quantum theory for computation. Quantum computers harness the properties such as superposition (quantum particles as combination of all possible states), entanglement (ability of quantum particles to correlate their measurement results with each other) and quantum interference (behaviour of a qubit, arising from superposition, which influences the probability of the qubit collapsing to one state or another upon measurement) for computing.

In conventional computers using semiconductors, electrons represent binary states (0 or 1), with eight bits covering numbers singly between 0 and 255. Quantum computing uses quantum bits (qubits) that are superpositions of 'spin up' and 'spin down' states, allowing them to represent all numbers between 0 and 255 simultaneously. This allows quantum computers to perform many calculations in parallel. Spintronic devices have the potential for quantum computing. The spins of the electrons are expected to play an important role in the quantum computers, as they can be used as qubits or quantum bit.

Hexagonal boron nitride, a material consisting of just a few monolayers of atoms, can be stacked to serve as the insulating layer in the capacitors used in superconducting qubits. Diamonds with nitrogen-vacancy (NV) centers, nanowires, including semiconductor, superconducting, and topological nanowires are also used in quantum computing. Metals like Al, Nb are used in used in superconducting circuits. Quantum dots materials like indium arsenide (InAs) and gallium arsenide (GaAs) are employed to produce single photons, which are crucial for quantum communication applications. Quantum computing requires very low temperatures to avoid decoherance so that quibts remains in their quantum states and prevents error rates.

2.7 Supercapacitors

Supercapacitors or ultra capacitors are energy storage devices having high-capacity, with a capacitance value much higher than solid-state capacitors but with lower voltage limits. Both capacitors and batteries are energy storage devices. The differences between capacitors and batteries are given in table 2.1.
In supercapacitors energy is stored electrostatically and electrochemically and bridges the gap

Table 2.1: Differences between battery and capacitors

Battery	Capacitor
Electrical energy is stored in the form of chemical energy	Electrical energy is stored in the form of electric field (electrostatically)
Low or limited cyclic life	Very large cyclic life
Energy density is very high	Energy density is low
Lower power density (release energy slowly)	Higher power density (release energy quickly)
High internal resistance	Very low internal resistance

between traditional capacitors and batteries. A capacitor contains two metal plates separated by a dielectric material whereas a super capacitor generally consists of electrodes, electrolyte, separator and collector.

Electrode: Materials with good conductivity and high stability like porous active carbon coating are generally used as the electrode material. Carbon based nanoparticles (eg. carbon nano tubes, graphene, carbon quantum dots etc.), metal oxide nanoparticles, nanowires, quantum dots, conducting polymer-based composites etc. are used as the electrode material in supercapacitors. Nanomaterials enhances the performance of the supercapacitors by increasing the capacitance, energy density, and cycling stability of the supercapacitors

Electrolyte: Capacity of the capacitor depends mainly on the dielectric constant of the electrolyte. Either solid or liquid can be used as the electrolyte. For commercial application solid electrolytes are preferred since they are leak-free and possess high ionic conductivity. Generally, a solvent mixed with conductive salts such as tetraalkylammonium or lithium salts acts as the solid electrolyte. Sulphuric acid, KOH solution etc can act as the liquid electrolyte.

Separator: Electrolyte membranes will act as the separator. It prevents short-circuiting between the electrodes but allows the electrolyte ions to pass through. Polymer-based and paper-based are found to be durable and economical.

Collector: Electrons are collected at the collector. Carbon fibre, metal like Al, Pt, Cu, Ni etc. are the commonly used collectors. Binders, such as polyvinylidene fluoride (PVDF) and polyacrylonitrile (PAN), act as adhesives to hold the active material and current collectors together.

Advantages of supercapacitor

- Longer cycling time and higher service life compared to the battery.
- High efficiency, higher charging rates and high power density.
- Low resistance which enables them to produce high load currents.
- Small sizes and lightweight which makes them easily installed in small areas.
- Environment friendly.

Disadvantages of supercapacitor

- Higher self discharge rate.
- Amount of energy stored per unit weight is considerably lower compared to batteries.
- Low voltage limits which demand serial connections to produce high voltage.

Applications of supercapacitors

- Used in hybrid buses in combination with the battery to increase battery life and decrease the size.
- Used in implantable devices and health monitoring devices to supply power.
- Used in solar-powered road and farming devices, street light portable traffic light system, wireless security camera etc.
- Used in wind turbines to supply power to the pitch control of blade.
- Memory devices in laptops, smartphones, tablets etc., are developed using supercapacitors.

2.8 Questions

2.8.1 Short answer questions

1. "Properties of nanomaterial differ from bulk material" rationalize the statement with suitable example.
2. Write a note on sol-gel method of nanoparticle synthesis.
3. Give any three applications of nanomaterials.
4. List any three properties of graphene.
5. Explain the electrical conductivity of graphene based on its structure.
6. List any three applications of carbon nano tubes.
7. What are carbon quantum dots?
8. What are fullerenes?
9. Discuss the structure and bonding in Buckminster fullerene.
10. How halogenated polymers quench fire?
11. Name and write the structure of any two fire retardant polymers.
12. What are conducting polymers? Draw the structure of polyacetylene and polyaniline.
13. Explain doping in conducting polymers.

14. Why conducting polymers like poly aniline and poly pyrrole can be doped only with p type dopents?
15. Show how polaron moves in a conducting polymer.
16. What are polaran and bipolaran?
17. Write the synthesis of polypyrrole.
18. Mention any three advantages of OLEDs over LED and LCD.
19. Why n region acts as emissive layer in contrast conventional inorganic LEDs.
20. Give any three advantages of dye sensitized solar cells.
21. What do you mean by spintronics.
22. List the materials used for quantum computing

2.8.2 Essay questions

1. Discuss the classification of nanomaterials with example.
2. Give the properties and application of nano materials.
3. Explain the sol-gel method and chemical reduction methods for the synthesis of nano particles.
4. Discuss the synthesis, properties and application of graphene.
5. Explain the classification of carbon nano tubes.
6. Describe the synthesis, properties and application of carbon nano tubes.
7. Discuss the properties and application of carbon quantum dots.
8. Explain the structural features and applications of fullerenes.
9. Explain the classification of conducting polymer.
10. What is meant by doping of polymers? Describe the different types of doping.
11. How is polyaniline synthesized? List any two properties and applications.
12. How is polypyrrole synthesized? List any two properties and applications.
13. What are OLEDs? Explain the construction and working of OLED with the help of a neat labelled sketch.
14. What are dye sensitized solar cells? Explain the construction and working of DSSC with the help of a neat labelled sketch.
15. Give short notes on materials used for spintronics and quantum computing.
16. Mention the advantages, disadvantages and applications of super capacitors.
17. Explain the materials used in supercapacitors.